

CellGalaxy

Human-in-the-loop Characterization of Tissue Microenvironments: From Intensity Averaging to Representation Learning

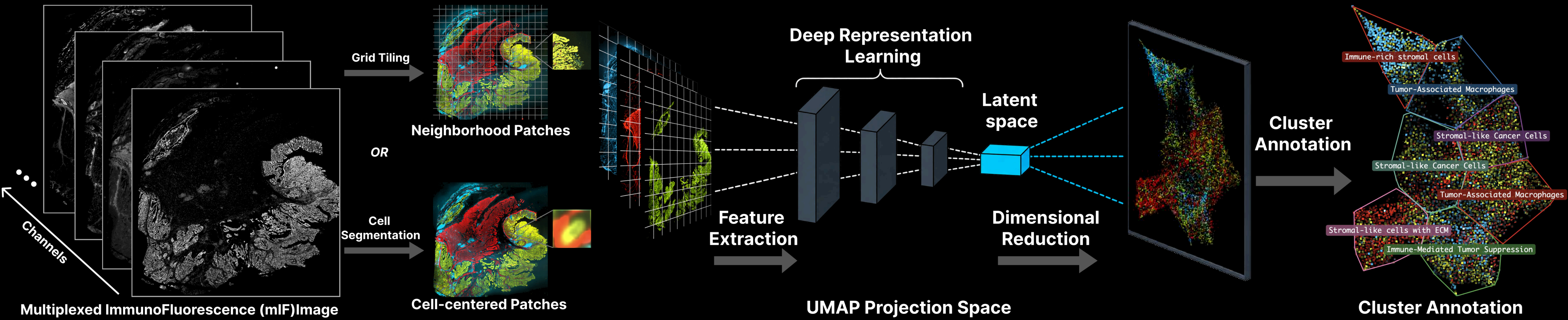
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Motivation. Multiplex immunofluorescence (mIF) imaging captures the complex spatial architecture of tumors at single-cell resolution but generates high-dimensional datasets that far exceed the limits of manual interpretation. Traditional methods summarize protein expression using features such as average intensity, preserving part of the key characteristics. To analyze such data at scale, deep vision models map tissue morphology and protein expression into lower-dimensional embeddings. While these embeddings organize millions of cells and image regions, they remain difficult to interpret, creating a bottleneck in biological discovery.

Approach. We present CellGalaxy, a scalable visual analysis approach for exploring and interpreting embeddings of multiplex biomedical image collections. CellGalaxy interactively renders embedding projections with millions of images and employs semantic zooming to adapt representations across levels of granularity. This enables navigation from spatial organization to detailed cellular patterns. Spatial and feature-based filters allow users to focus on regions of interest, examine image features, compare spatial compositions, and link back to the original tissue context across samples. To accelerate biological interpretation, CellGalaxy uses Large Language Models (LLMs) to enrich embedding regions with automated contextual descriptions, featuring an interactive workflow for users to proofread, accept, or decline the results.

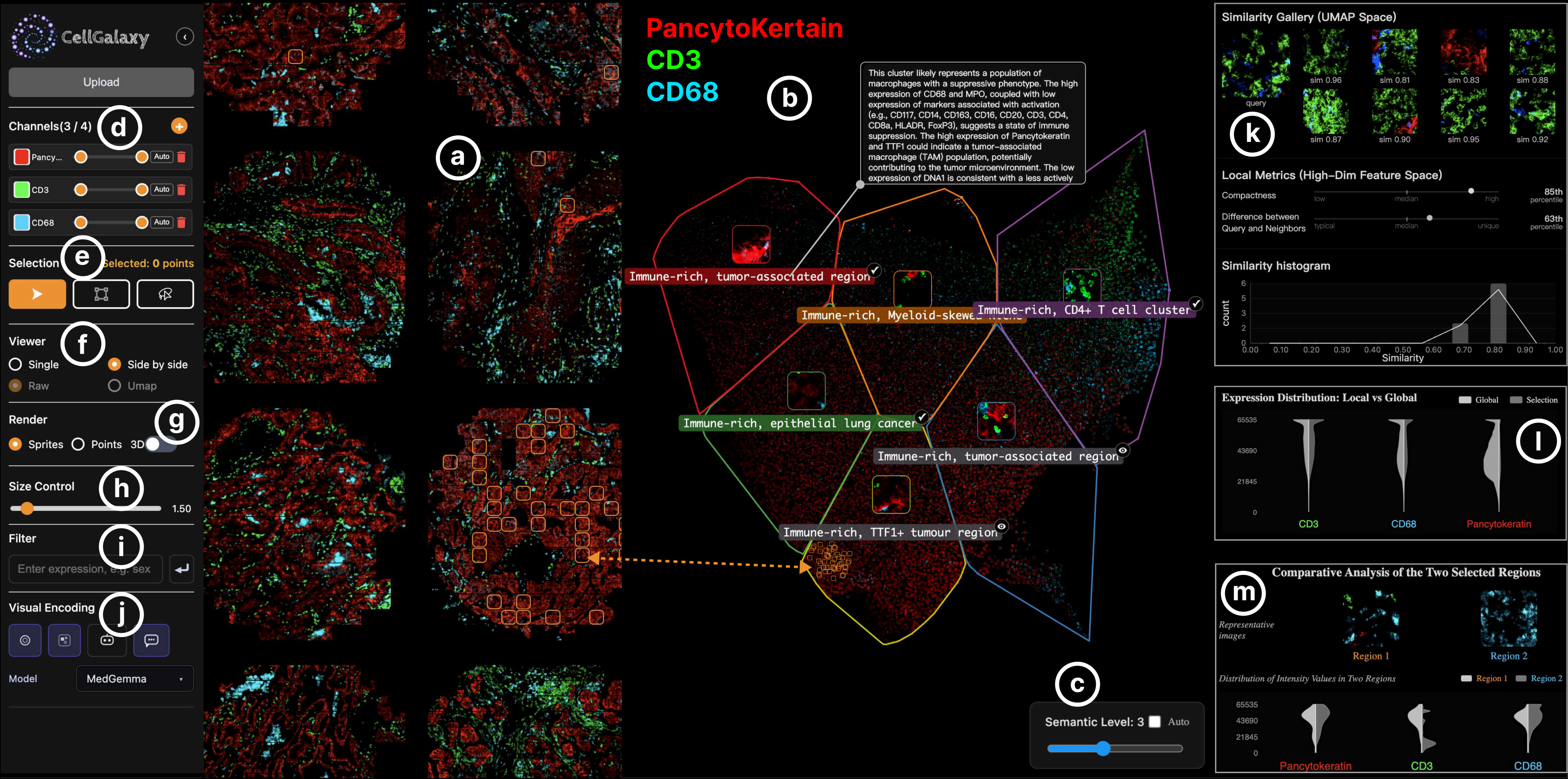
Evaluation. We evaluate CellGalaxy through scalability experiments, expert review of LLM-generated descriptions, three case studies with experts on different cancer tissues, and a hands-on study with 12 users to assess the system's usability and analytical utility. The results confirm that CellGalaxy enables scalable embedding exploration, yielding biologically meaningful insights into both cellular heterogeneity and tissue organization.

Data preprocessing pipeline



Data preprocessing pipeline. Multiplexed Immunofluorescence (mIF) image consisting of multi-channels; patch generation through either fixed-size tiling or cell segmentation; feature encoding using deep models; projection of high-dimensional embeddings into 2D/3D UMAP space, where groups of similar patches form clusters that support multi-scale exploration; annotation of clusters with textual labels.

CellGalaxy interface



CellGalaxy interface overview. (a–b) Spatial and UMAP embedding views of image patches (tiles or single-cell crops); (c) semantic zooming control; (d) multi-channel marker visualization; (e) linked selection panel; (f) single and side-by-side viewer; (g) rendering modes with 2D/3D embeddings; (h) sprite size control; (i) metadata filtering; (j) cluster-level visual encoding; (k) similarity exploration in feature space; (l) marker expression distributions; (m) comparative analysis across regions.



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